

BUS43-4

International Economics & Globalization

Trade, exchange rates, and the politics of open borders

Applying the micro/macro toolkit to open economies.

Albert School — 22 hours

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1 Comparative advantage and the distributional theorems

1.1 Ricardian comparative advantage

A country has a *comparative advantage* in the good it produces at lower opportunity cost, even if it has an absolute disadvantage in both goods. With one factor (labour) and constant unit labour requirements a_{Li} for good i , country Home exports the good in which its relative unit labour requirement is lower.

Definition 1 (Comparative advantage) With two goods 1, 2 and two countries Home and Foreign, Home has a comparative advantage in good 1 when

$$\frac{a_{L1}}{a_{L2}} < \frac{a_{L1}^*}{a_{L2}^*},$$

where starred terms are Foreign's unit labour requirements. The opportunity cost of good 1 in terms of good 2 is a_{L1}/a_{L2} .

Worked example — Gains from trade. Suppose Home needs 1 hour per unit of cloth and 2 hours per unit of wine, Foreign needs 6 and 3. Home's opportunity cost of cloth is 1/2 wine; Foreign's is $6/3 = 2$ wine. Home has the comparative advantage in cloth, Foreign in wine. Both gain by specialising and trading at any relative price between the two autarky price ratios 1/2 and 2.

The gains from trade are realised because the world relative price settles between the two countries' autarky relative prices; each country consumes outside its own production possibility frontier.

1.2 Heckscher–Ohlin

The Heckscher–Ohlin model derives comparative advantage from factor endowments. A country exports the good that uses intensively the factor with which it is relatively abundant.

Theorem 2 (Heckscher–Ohlin) A country relatively abundant in a factor exports the good whose production uses that factor intensively.

1.3 Stolper–Samuelson

Trade liberalisation changes goods prices, and through them factor rewards.

Theorem 3 (Stolper–Samuelson) A rise in the relative price of a good raises the real return of the factor used intensively in that good and lowers the real return of the other factor.

Consequently, in a country abundant in skilled labour, opening to trade raises the return to skilled labour and lowers the return to the scarce factor (unskilled labour). The model predicts that trade liberalisation creates winners and losers *within* a country, not just across countries. This prediction is tested against the wage stagnation of the rich-country middle class: the scarce factor in advanced economies — less-skilled labour — is predicted to lose relative to capital and skilled labour.

2 New trade theory

The Ricardian and Heckscher–Ohlin models explain *inter-industry* trade between dissimilar countries. Much observed trade is *intra-industry* trade between similar countries (e.g. cars for cars). New trade theory (Krugman) explains it through:

- **Internal economies of scale:** average cost falls with firm output, so the world supports only a limited number of varieties; each country specialises in some varieties.
- **Product differentiation** under monopolistic competition: consumers value variety, so trade expands the range of available varieties and lowers prices.
- **Economic geography:** the interaction of scale economies and transport costs concentrates production in particular locations.

Intra-industry trade is the empirical face of modern globalization: gains arise from scale and variety rather than from comparative-advantage specialisation, and the distributional disruption is smaller because trade is between similar factor endowments.

3 Tariff welfare accounting

Consider a small importing country facing a fixed world price p_w . A specific tariff t raises the domestic price to $p_w + t$. Domestic quantity supplied rises and quantity demanded falls; imports shrink.

Definition 4 (Welfare components of a tariff) Relative to free trade, a tariff t in a small country produces:

- a *loss of consumer surplus* (the area between demand and price over the higher price), area $a + b + c + d$ in the standard decomposition;
- a *gain in producer surplus*, area a ;

- *fiscal revenue* $t \times M_t$, where M_t is the post-tariff import volume, area c ;
- a *deadweight loss* equal to the production-distortion triangle b plus the consumption-distortion triangle d .

Net national welfare change is $-(b + d)$.

Common pitfall For a *small* country the terms of trade are fixed, so a tariff is unambiguously welfare-reducing: the net effect is the deadweight loss $-(b + d)$.

For a *large* country the tariff lowers the world price it pays (a *terms-of-trade* gain, the rectangle e accruing from foreign exporters). Net welfare is then $e - (b + d)$, which can be positive: the optimal tariff balances the terms-of-trade gain against the deadweight loss. The gain comes at the expense of trading partners.

4 Non-tariff barriers and the case for protection

4.1 Instruments

- **Quota**: a quantitative limit on imports. It raises the domestic price like a tariff, but the rectangle that would be tariff revenue becomes *quota rents* captured by licence holders (or foreign exporters under a voluntary export restraint).
- **Export subsidy**: a per-unit payment to exporters. It raises the domestic price, lowers the world price, transfers to producers, and produces a deadweight loss plus, for a large country, a terms-of-trade *loss*.
- **Other non-tariff barriers**: standards, licensing, local-content rules and administrative procedures that raise the cost of imports.

4.2 Protectionist arguments

- **Infant industry**: temporary protection lets a young industry reach the scale or experience at which it becomes competitive. Valid only if the industry eventually survives unprotected and the market failure (capital-market or learning externality) is real.
- **Reciprocity**: protection as leverage to obtain reductions abroad; risks retaliation and trade wars.
- **National security**: protection of strategically essential sectors. Hard to bound, since many sectors can be labelled essential.

Each argument must be weighed against the deadweight loss it imposes; the cost-benefit test is whether the claimed gain exceeds the welfare triangles.

4.3 The WTO

The World Trade Organization administers multilateral trade rules: most-favoured-nation (non-discrimination), national treatment, binding tariff ceilings, and a dispute-settlement mechanism. Its *limits*: the Doha round stalled, the Appellate Body has been paralysed since 2019 (blocked appointments), and the rules constrain but do not prevent the contemporary turn to industrial policy and unilateral tariffs.

5 Contemporary trade-policy stress tests

Episode	Analytical reading
US–China tariffs 2018–2025	Large-country tariffs on Chinese imports. Tariff welfare accounting: US consumers and import-using firms lose (price rises borne largely domestically), protected US producers gain, the Treasury collects revenue; net deadweight loss. Retaliation reduces US exports. Winners: shielded import-competing sectors; losers: US consumers, exporters facing retaliation, integrated supply chains.
US Inflation Reduction Act (IRA)	Production subsidies and local-content conditions for clean energy. A domestic subsidy with terms-of-trade and trade-diversion effects: it attracts investment to the US, disadvantages foreign (incl. EU) producers, and risks a subsidy race. Winners: US clean-tech producers and workers; losers: foreign competitors, foreign taxpayers funding countervailing subsidies.
EU Carbon Border Adjustment Mechanism (CBAM)	A border charge on the carbon content of imports, equalising the carbon price faced by domestic and foreign producers. Reads as a tariff calibrated to an externality (carbon leakage). Winners: EU producers under the ETS, the climate objective; losers: carbon-intensive exporters to the EU, who bear the charge.

6 Balance of payments

The balance of payments records all transactions between residents and the rest of the world over a period.

Definition 5 (Accounts and identity)

- **Current account (CA):** trade in goods and services, primary income (factor income), and secondary income (transfers).
- **Capital account:** capital transfers and acquisition of non-produced non-financial assets (small).
- **Financial account (FA):** net acquisition of foreign assets minus net incurrence of foreign liabilities (direct, portfolio, other investment, reserves).

By construction (sign conventions of the IMF standard),

$$CA + \text{capital account} - FA + \text{errors and omissions} = 0.$$

A current-account *deficit* means a country spends more than it produces and must be financed by net capital inflows (rising net foreign liabilities). A *surplus* is the mirror image: net lending to the rest of the world. The identity links the current account to the saving–investment balance, $CA = S - I$.

7 Exchange-rate determination

7.1 Purchasing power parity (long run)

Definition 6 (PPP) *Absolute PPP*: the nominal exchange rate equates price levels,

$$E = \frac{P}{P^*},$$

with E units of home currency per unit of foreign currency, P , P^* the price levels. *Relative PPP*: the rate of depreciation equals the inflation differential,

$$\frac{\Delta E}{E} = \pi - \pi^*.$$

PPP is a long-run anchor: the real exchange rate $Q = EP^*/P$ is mean-reverting but deviates persistently in the short run.

7.2 Uncovered interest parity (short run)

Definition 7 (UIP) Capital mobility equalises expected returns across currencies:

$$i = i^* + \frac{E^e - E}{E},$$

i.e. the domestic interest rate equals the foreign rate plus the expected depreciation of the home currency. Equivalently, the expected appreciation of a currency offsets its interest-rate disadvantage.

Worked example — UIP numerical application. If $i = 4\%$ and $i^* = 1\%$, UIP requires an expected home-currency depreciation of about 3% over the horizon. A higher domestic rate attracts capital inflows and causes an immediate appreciation, with subsequent expected depreciation restoring parity.

8 The Mundell–Fleming trilemma

Theorem 8 (Impossible trinity) A country cannot simultaneously have all three of: a *fixed exchange rate*, *free capital mobility*, and an *independent monetary policy*. At most two can be chosen.

- **Eurozone:** members fixed the exchange rate (single currency) and kept capital mobility, surrendering national monetary policy to the ECB.

- **China (RMB)**: long combined a managed exchange rate with monetary autonomy by restricting capital mobility through capital controls.
- **Emerging economies**: typically choose a floating rate plus capital mobility to retain monetary independence, accepting exchange-rate volatility.

9 Currency crises

Generation	Mechanism
First generation (Krugman)	A fixed exchange rate is incompatible with persistent fiscal deficits monetised by the central bank. Reserves fall steadily; a speculative attack exhausts them at a predictable point and the peg collapses. The crisis is the result of inconsistent fundamentals.
Second generation (Obstfeld)	Self-fulfilling crises with multiple equilibria. The government trades off the cost of defending the peg against the cost of abandoning it; if markets expect devaluation, defending becomes too costly and the expectation is validated, even with sustainable fundamentals.
Third generation	Crises driven by balance-sheet effects, banking fragility, and currency mismatches: firms and banks borrow in foreign currency; a depreciation raises the domestic value of their debt, causing insolvency, capital flight, and further depreciation.

1997 Asian crisis: large foreign-currency corporate and bank debt, fixed or quasi-fixed pegs, and sudden capital-flow reversals — primarily a third-generation balance-sheet crisis. **2001 Argentina**: a currency board fixed the peso one-to-one to the dollar; fiscal stress and an over-valued real rate under $r > g$ debt dynamics made the peg unsustainable, ending in abandonment and default — first- and second-generation features.

10 Global value chains

Modern production is fragmented across borders. The OECD’s *Trade in Value Added* (TiVA) database measures the value each country adds to a final good, rather than gross trade flows, correcting the double-counting in gross statistics and revealing where value is actually created.

A *dependency map* traces, for a critical input, which countries and firms supply it and how concentrated that supply is. *Supply-chain stress* arises when a concentrated node is disrupted:

- **COVID-19**: lockdowns and demand swings disrupted just-in-time chains (semiconductors, shipping), revealing the fragility of single-sourced inputs.
- **Ukraine war**: disruption to energy, grain, and key metals exposed European dependence on Russian gas and concentrated commodity suppliers.
- **Semiconductor export controls**: dependence on a few chokepoints (advanced fabrication, lithography) turned a supply chain into an instrument of policy.

The diagnosis identifies concentrated, hard-to-substitute nodes as the points of geopolitical risk.

11 The distributional politics of globalization

Definition 9 (Elephant curve (Milanovic)) Plotting real income growth (1988–2008) against the global income distribution percentile traces a shape resembling an elephant:

- large gains for the *emerging-economy middle class* (around the median of the global distribution, chiefly Asia);
- near-stagnation for the *rich-country middle and lower-middle class* (around the 80th–90th global percentiles);
- large gains for the *global top 1%*.

The curve connects to contemporary trade politics: the relative stagnation of the rich-country middle class — consistent with the Stolper–Samuelson prediction for the scarce factor in advanced economies — underlies the political backlash against globalization, even as global poverty fell and inequality *between* countries narrowed.

12 Geopolitical reconfigurations

- **Global imbalances:** persistent surpluses (Germany, China) mirror persistent deficits (United States). By the identity $CA = S - I$, surpluses reflect saving in excess of investment; the US deficit is financed by capital inflows and sustained by the dollar’s reserve role.
- **Reshoring and friendshoring:** relocating production home or to allied countries reduces dependency risk but raises costs — sacrificing comparative-advantage and scale gains for resilience.
- **Economic weaponisation:** the use of economic interdependence as leverage — financial sanctions and exclusion from payment systems, the primacy of the dollar as a chokepoint, and semiconductor export controls. These convert the networks of globalization into instruments of geopolitical power.